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Indian Institute of Technology Dharwad

CS101C: Introduction to Programming (Using C)

Instructors: Prof. Aikta Arya and Prof. Konjengbam Anand

CSE Department

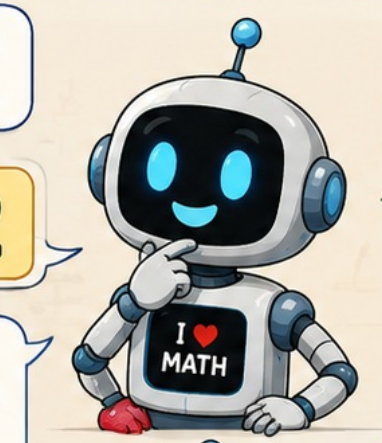
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Week 2

Today's Class (10/8/2026)

OPERATORS – THE ACTION MAKERS!

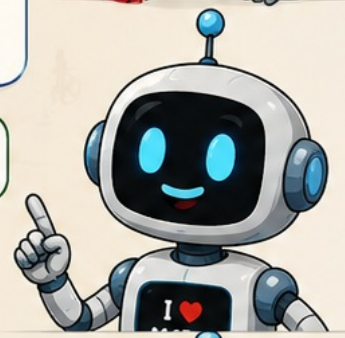




Add them.

8	+	2	=	10
First Number	Operator (Add)	Second Number		Result

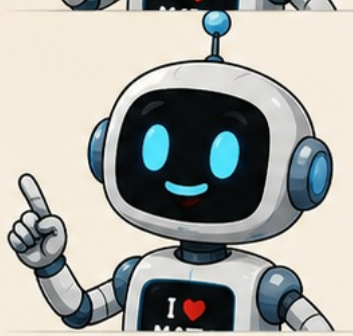
That's 10!



Subtract them.

8	−	2	=	6
First Number	Operator (Subtract)	Second Number		Result

That's 6!



Multiply them.

8	×	2	=	16
First Number	Operator (Multiply)	Second Number		Result

That's 16!



Operators tell the computer what action to perform.

8 → + → 2 → 10

Number — Operator — Number — Result

Operators

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- Arithmetic $(+, -, *, /, \%)$
- Relational $(==, !=, >, <, >=, <=)$
- Assignment $(+=, -=, *=, /=, \%, <<=, >>=, \&=, ^=, |=)$ (later)
- Increment / Decrement $(++, --)$
- Special: ternary, sizeof

Arithmetic Operators – Example Program

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```c
#include <stdio.h>

int main()
{
    int a = 10, b = 3;

    printf("a + b = %d\n", a + b);
    printf("a - b = %d\n", a - b);
    printf("a * b = %d\n", a * b);
    printf("a / b = %d\n", a / b);
    printf("a %% b = %d\n", a % b);

    return 0;
}
```

Output

Console Output

```
a + b = 13
a - b = 7
a * b = 30
a / b = 3
a % b = 1
```

Relational Operators – Example Program

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```c
#include <stdio.h>

int main()
{
    int a = 10, b = 3;

    printf("a == b : %d\n", a == b);
    printf("a != b : %d\n", a != b);
    printf("a > b  : %d\n", a > b);
    printf("a < b  : %d\n", a < b);
    printf("a >= b : %d\n", a >= b);
    printf("a <= b : %d\n", a <= b);

    return 0;
}
```

Output

Console Output

```
a == b :  0
a != b :  1
a > b :  1
a < b :  0
a >= b :  1
a <= b :  0
```

Assignment Operators – Example Program

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```c
#include <stdio.h>
int main()
{
    int x, a = 10, b = 3;
    x = a;
    x += b;
    printf("x += b : %d\n", x);
    x = a;
    x -= b;
    printf("x -= b : %d\n", x);
    x = a;
    x *= b;
    printf("x *= b : %d\n", x);
    x = a;
    x /= b;
    printf("x /= b : %d\n", x);
    x = a;
    x %= b;
    printf("x %%= b : %d\n", x);
    return 0;
}
```

Output

Console Output

```
x += b :  13
x -= b :  7
x *= b :  30
x /= b :  3
x %= b :  1
```

Increment / Decrement Operators – Example Program


```c
#include <stdio.h>

int main()
{
    int a = 10;
    int x = a;

    printf("x++ : %d\n", x++);
    printf("++x : %d\n", ++x);
    printf("x-- : %d\n", x--);
    printf("--x : %d\n", --x);

    return 0;
}
```

Output

Console Output

```
x++ :  10
++x :  12
x-- :  12
--x :  10
```

Special Operators – Example Program

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```c
#include <stdio.h>

int main()
{
    int a = 10, b = 3;

    printf("Size of a: %zu\n", sizeof(a));

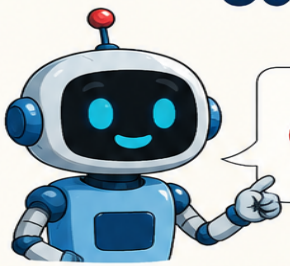
    int result = (a = b + 2, a * 2);
    printf("Comma operator result: %d\n", result);

    int max = (a > b) ? a : b;
    printf("max(a, b): %d\n", max);

    return 0;
}
```

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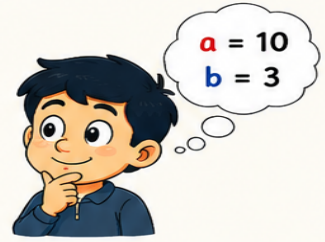
Comma Operator in Action!



Let's see how
(a = b + 2, a * 2)
works step by step!

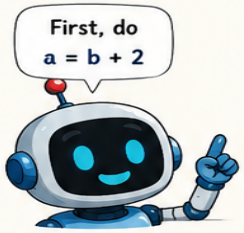
Initially

a = 10
b = 3



(a = b + 2, a * 2)

Two expressions separated by a comma. They are evaluated from left to right.

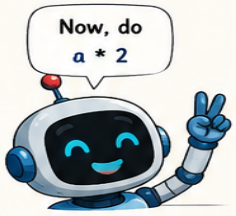


Step 1: Evaluate the first expression

a = b + 2 → a = 3 + 2, a = 5

After Step 1

a = 5
b = 3

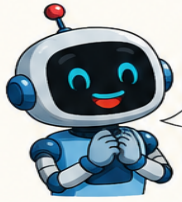


Step 2: Evaluate the second expression

a * 2 → 5 * 2 = 10

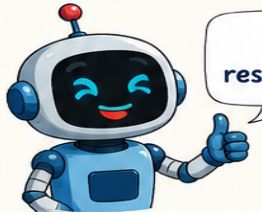
After Step 2

a = 5
b = 3



The value of the last expression (a * 2) is returned.

10



Final Result

result = 10



Ternary Operator (? :) – Find the Maximum

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1 Initially



```
int a = 10;
int b = 3;
```

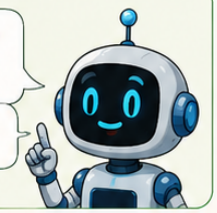
```
int max = (a > b) ? a : b;
```

Format: (condition) ? value_if_true : value_if_false



2 Step 1: Evaluate the condition

a > b → 10 > 3 → TRUE



3 Step 2: Condition is TRUE



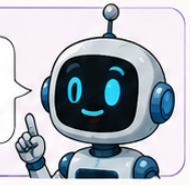
(a > b) ? a : b

TRUE → ✓ Taken: a; ✗ Not taken: b

Take a → a = 10

4 Step 3: Assign the selected value to max

max = 10 → max 10



5 Output

```
printf("max(a, b): %d\n", max);
```

Output on screen: max(a, b): 10



Ternary operator is equivalent to:

```
if (a > b)    max = a;    else    max = b;
```

Summary

It checks the condition.
If TRUE → choose a, else → choose b.

Output

Console Output

```
Size of a:  4
Comma operator result:  10
max(a, b):  5
```

Today's Class (12/8/2026)

Operators

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- Arithmetic $(+, -, *, /, \%)$
- Relational $(==, !=, >, <, >=, <=)$
- Assignment $(+=, -=, *=, /=, \%, <<=, >>=, \&=, ^=, |=)$ (later)
- Increment / Decrement $(++, --)$
- Special: ternary, sizeof
- Logical (&&, ||, !)

Logical Operators – Example Program

- In C, zero means **FALSE**; anything non-zero means **TRUE**.

```c
#include <stdio.h>

int main()
{
    int a = 10, b = 3;

    printf("a && b: %d\n", a && b);
    printf("a || b: %d\n", a || b);
    printf("!a: %d\n", !a);
    printf("!b: %d\n", !b);

    return 0;
}
```

Output

Console Output

```
a && b:  1
a || b:  1
!a:  0
!b:  0
```

Sizeof Operator – Example Program

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- sizeof looks like a function, but it is actually an unary operator.

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int a=10;
6      float b=10.6;
7      char c='A';
8      double d=10.2;
9      short e= 1;
10     long f = 12;
11     printf("size of int a=  %zu byte(s)\n", sizeof(a));
12     printf("size of float b=  %zu byte(s)\n", sizeof(b));
13     printf("size of char c=    %zu byte(s)\n", sizeof(c));
14     printf("size of double d=   %zu byte(s)\n", sizeof(d));
15     printf("size of short e=   %zu byte(s)\n", sizeof(e));
16     printf("size of long f=   %zu byte(s)\n", sizeof(f));
17     return 0;
18 }
```

Output

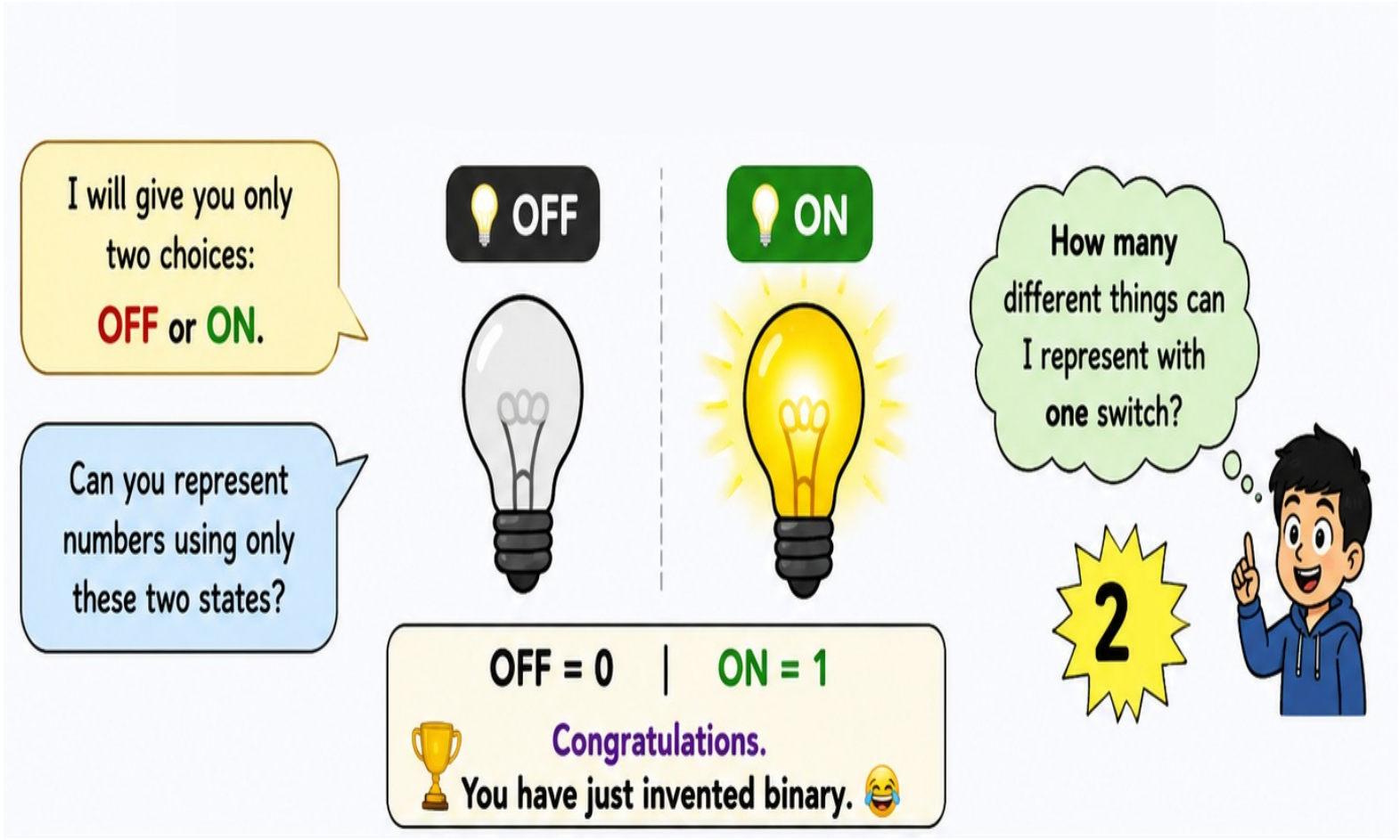
Console Output

```
size of int a= 4 byte(s)
size of float b= 4 byte(s)
size of char c= 1 byte(s)
size of double d= 8 byte(s)
size of short e= 2 byte(s)
size of long f= 4 byte(s)
```

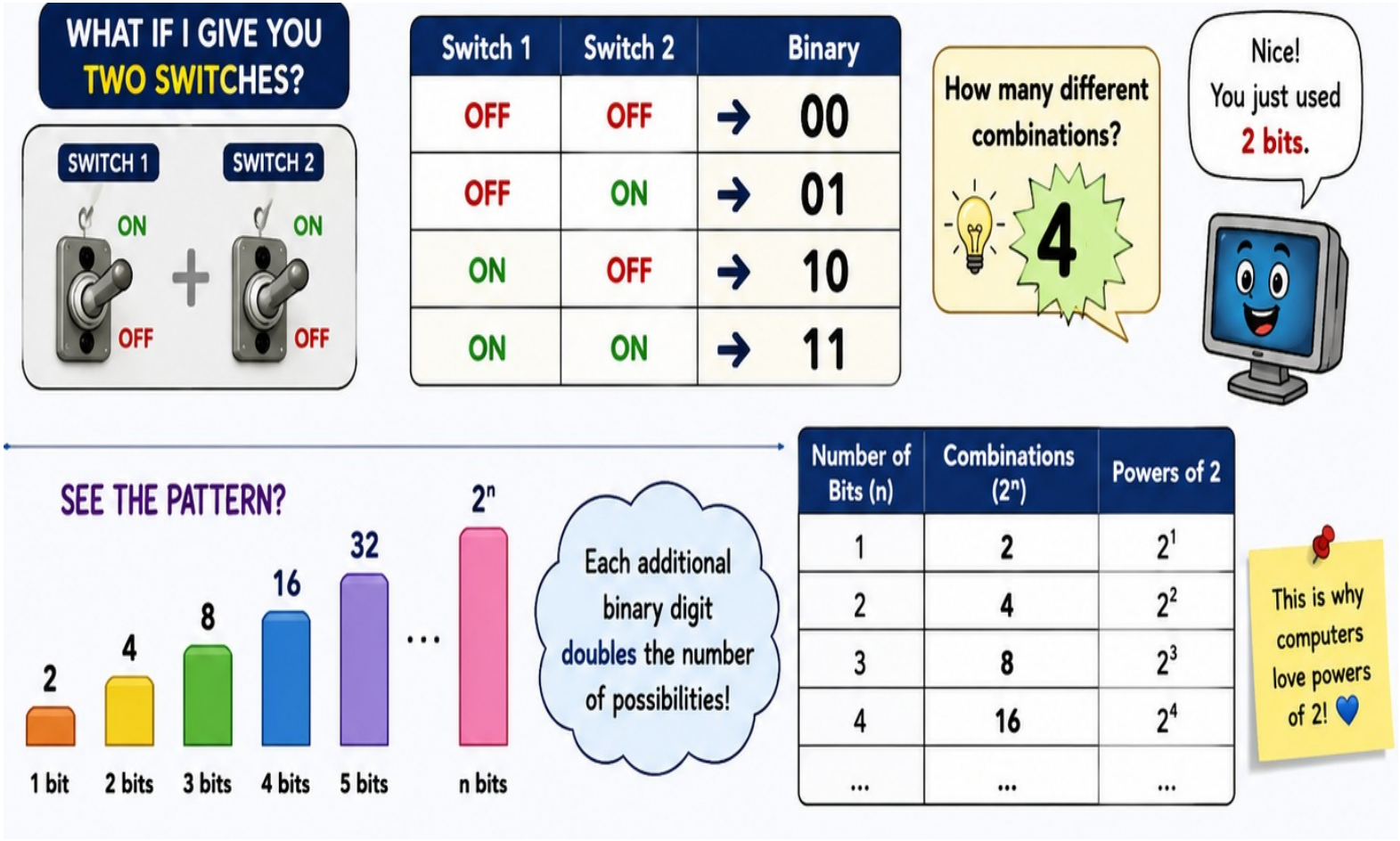
Common Format Specifiers

Specifier	Data Type	Function
%d	int	printf() / scanf()
%hd	short int	printf() / scanf()
%ld	long int	printf() / scanf()
%f	float	printf() / scanf()
%lf	double	scanf()
%c	char	printf() / scanf()
%zu	size_t	printf()

Number Bases



Number Bases



ONE NUMBER — DIFFERENT LANGUAGES

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Number System	Representation	Why?
Decimal	26_{10}	Humans normally use it
Binary	11010_2	Natural for digital hardware
Hexadecimal	$1A_{16}$	Compact way to write binary
Octal	32_8	Another compact representation

Same number. Different representation.

Number Bases

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- We use decimal (base-10), Computers use binary (base-2).
- Binary is difficult to read. So, we use Hexadecimal (base-16).
- Octal (base-8) is the other popular number format.

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NUMBER BASES – HEXADECIMAL

01 Hexadecimal uses 16 digits: 0 to 9 and A to F.
A to F represent decimal numbers 10 to 15.



Hexadecimal makes it easier to represent binary data in a shorter form!

Hexadecimal Digit	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal Equivalent	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

02 A digit in hexadecimal needs 4 bits. Therefore, a byte of information (8 bits) represents two hexadecimal digits.

1 Hexadecimal Digit = 4 Bits

Example: A

1	0	1	0

4 bits

Example: F

1	1	1	1

4 bits

1 Byte (8 Bits) = 2 Hexadecimal Digits

Example: 8 Bits (1 Byte)

1	0	1	0	1	1	0	1

8 bits

= 2 Hex Digits

A	D



A single hex digit (4 bits) can represent 16 values (0 to 15).

Binary (8 bits)	Hexadecimal
0000 0000	00
0000 0001	01
1010 1101	AD
1111 1111	FF

Using hexadecimal reduces the number of digits by half compared to binary, making data easier to read and write.

Number Bases- Octal

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NUMBER BASES – OCTAL

Octal is a base-8 number system.

01 What is Octal?

- Octal uses 8 digits: 0 to 7.
- Each digit in octal represents values from 0 to 7 (in decimal).

Octal Digit	0	1	2	3	4	5	6	7
Decimal Equivalent	0	1	2	3	4	5	6	7

02 How Many Bits Does One Octal Digit Need?

- Since $2^3 = 8$, one octal digit needs 3 bits.
- Therefore, each octal digit can represent values 0 to 7.

1 Octal Digit = 3 Bits

b_2	b_1	b_0
4	2	1

Values: 0 to 7

Bits vs Octal Digits

Bits	Octal Digits (Required)
3 bits	1
6 bits	2
8 bits	up to 3
9 bits	3
12 bits	4

Number Bases conversion — Decimal to Binary

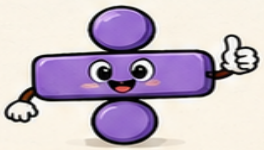


1. Divide integer by 2, note down remainder
2. Repeat step 1 till you get a zero in the quotient
3. Write down the remainders noted from last to first.

Number Bases conversion — Decimal to Binary

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1 Divide integer by 2, note down remainder.



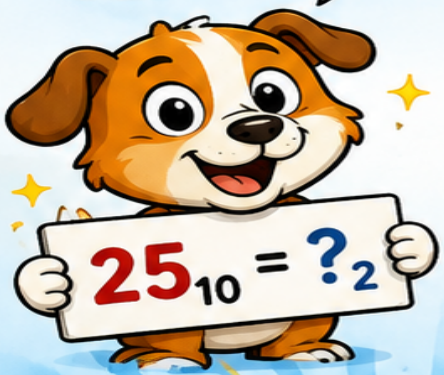
2 Repeat step 1 till you get a zero in the quotient.



3 Write down the remainders noted from last to first.



Let's convert **25** to binary!



Step	Divide by 2	Quotient	Remainder
1	25 ÷ 2	12	1
2	12 ÷ 2	6	0
3	6 ÷ 2	3	0
4	3 ÷ 2	1	1
5	1 ÷ 2	0	1

Collect remainders from bottom to top!

1 0 0 1 1

Why does it work?

In binary, each position represents a power of 2: 1, 2, 4, 8, 16, ...

Collecting remainders gives the coefficients of those powers!

16	8	4	2	1
1	1	0	0	1

= 16 + 8 + 0 + 0 + 1
= 25

So, $25_{10} = 11001_2$

Easy, right? Just divide by 2 and read from bottom to top!



Number Bases conversion — Decimal to Binary



Example1: Convert 13_{10} to binary.

Number Bases conversion — Decimal to Binary

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Example1: Convert 13_{10} to binary.

Repeatedly divide by 2:

$$\begin{aligned} 13 \div 2 &= 6 \quad \text{remainder } 1 \\ 6 \div 2 &= 3 \quad \text{remainder } 0 \\ 3 \div 2 &= 1 \quad \text{remainder } 1 \\ 1 \div 2 &= 0 \quad \text{remainder } 1 \end{aligned}$$

Read the remainders from bottom to top:

$$\boxed{1101}$$

Therefore,

$$\boxed{13_{10} = 1101_2}$$

Number Bases conversion — Decimal to Binary

Example2: Convert 500_{10} to binary.

Number Bases conversion — Decimal to Binary

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Example2: Convert 500_{10} to binary.

$$
\begin{aligned}
500 \div 2 &= 250 \quad \text{remainder } 0 \\
250 \div 2 &= 125 \quad \text{remainder } 0 \\
125 \div 2 &= 62 \quad \text{remainder } 1 \\
62 \div 2 &= 31 \quad \text{remainder } 0 \\
31 \div 2 &= 15 \quad \text{remainder } 1 \\
15 \div 2 &= 7 \quad \text{remainder } 1 \\
7 \div 2 &= 3 \quad \text{remainder } 1 \\
3 \div 2 &= 1 \quad \text{remainder } 1 \\
1 \div 2 &= 0 \quad \text{remainder } 1
\end{aligned}
$$

Read the remainders from bottom to top:

$$\boxed{111110100}$$

Therefore,

$$\boxed{500_{10} = 111110100_2}$$

Number Bases conversion — Binary to Decimal

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1. Consider the binary digits. Start from the right-most position and move towards left.
2. Multiply each binary digit with power-of-two. Starting from zero (right-most) and increment the power for each digit as you move towards left.
3. Sum all the results obtained in step 2.

Number Bases conversion — Binary to Decimal

Example 1: Convert 10110110_2 to decimal.

Number Bases conversion — Binary to Decimal

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Example1: Convert 10110110_2 to decimal.

Write the powers of 2:

1	0	1	1	0	1	1	0
128	64	32	16	8	4	2	1

$$10110110_2 = 128 + 32 + 16 + 4 + 2$$
$$= 182$$

$$\boxed{10110110_2 = 182_{10}}$$

Number Bases conversion — Binary to Decimal

Example2: Convert 1011011010_2 **to decimal.**

Number Bases conversion — Binary to Decimal

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Example2: Convert 1011011010_2 to decimal.

Write the powers of 2:

1	0	1	1	0	1	1	0	1	0
512	256	128	64	32	16	8	4	2	1

$$\begin{aligned} 1011011010_2 &= 1(512) + 0(256) + 1(128) + 1(64) + 0(32) \\ &\quad + 1(16) + 1(8) + 0(4) + 1(2) + 0(1) \\ &= 512 + 128 + 64 + 16 + 8 + 2 \\ &= 730 \end{aligned}$$

$$\boxed{1011011010_2 = 730_{10}}$$

Today's Class (14/8/2026)

In Today's class we'll cover

- Arithmetic $(+, -, *, /, \%)$
- Relational $(==, !=, >, <, >=, <=)$
- Assignment $(+=, -=, *=, /=, \%, <<=, >>=, \&=, \hat{}=, |=)$ (later)
- Increment / Decrement $(++, --)$
- Special: ternary, sizeof
- Logical (&&, ||, !)
- Bitwise $(\&, |, \hat{}, \sim, <<, >>)$
- Binary to Decimal to Hexadecimal (and combinations)

Number Bases conversion — Hexadecimal to Binary

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

1. A in binary = 1010
2. AB in binary = 1010 1011.
3. But when we write we use a notation: 0xA, 0xAB (read as "zero ex A B")

Exercise: convert 0xDEFC to binary
Exercise: convert 0xAB4 to binary

Number Bases conversion — Hexadecimal to Binary

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1. A in binary = 1010
2. AB in binary = 1010 1011.
3. But when we write we use a notation: 0xA, 0xAB (read as "zero ex A B")

Exercise: convert 0xDEFC to binary 1101 1110 1111 1100
Exercise: convert 0xAB4 to binary 1010 1011 0100

Number Bases conversion — Binary to Hexadecimal

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1. Starting from right most digit, group the binary digits in sets of 4 consecutive digits.
2. Consider the binary number formed from 4 digits in each set.
3. Map the binary number to hexadecimal.

Example:

$$1001\ 1011\ 1011$$

$$\underbrace{1001}_{9}\quad\underbrace{1011}_{B}\quad\underbrace{1011}_{B}$$

Therefore,

$$\boxed{100110111011_2 = 9BB_{16}}$$

Number Bases conversion — Binary to Hexadecimal

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Example: Convert 10011011011_2 **to hexadecimal.**

Step 1: Group the binary digits into groups of 4

$$10011011011$$

$$\boxed{0100 \quad 1101 \quad 1011}$$

Step 2: Convert each group

$$\underbrace{0100}_{4} \quad \underbrace{1101}_{D} \quad \underbrace{1011}_{B}$$

$$\boxed{10011011011_2 = 4DB_{16}}$$

Number Bases conversion — Hexadecimal to Decimal

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- Consider the Hexadecimal digits. Start from the right-most position and move towards left.
- Multiply each binary digit with power-of-sixteen. Starting from zero (right-most) and increment the power for each digit as you move towards left.
- Sum all the results obtained in step 2.

Example: What is the decimal equivalent of **0x4BD**?

Number Bases conversion — Hexadecimal to Decimal

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- Consider the Hexadecimal digits. Start from the right-most position and move towards left.
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- Sum all the results obtained in step 2.

Example: What is the decimal equivalent of **0x4BD**?

$$\begin{aligned} 4BD_{16} &= 4(16^2) + B(16^1) + D(16^0) \\ &= 4(256) + 11(16) + 13(1) \\ &= 1024 + 176 + 13 \\ &= \boxed{1213} \end{aligned}$$

$$\boxed{0x4BD = 1213_{10}}$$

Number Bases conversion — Hexadecimal to Decimal

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भारतीय प्रौद्योगिकी संस्थान धारवाड
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Example: What is the decimal equivalent of **0x2F7**?

Number Bases conversion — Hexadecimal to Decimal

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भारतीय प्रौद्योगिकी संस्थान धारवाड
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Example: What is the decimal equivalent of **0x2F7**?

$$
\begin{aligned}
2F7_{16} &= 2(16^2) + F(16^1) + 7(16^0) \\
&= 2(256) + 15(16) + 7(1) \\
&= 512 + 240 + 7 \\
&= \boxed{759}
\end{aligned}
$$

$$\boxed{0x2F7 = 759_{10}}$$

Bitwise Operators — Example Program

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भारतीय प्रौद्योगिकी संस्थान धारवाड
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```c
#include <stdio.h>

int main()
{
    int a=10, b=3;

    printf("a & b: %d\n", a & b);
    printf("a | b: %d\n", a | b);
    printf("a ^ b: %d\n", a ^ b);
    printf("~a: %d\n", ~a);
    printf("a << 1: %d\n", a << 1);
    printf("a >> 1: %d\n", a >> 1);

    return 0;
}
```

Bitwise Operators – Summary

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Operation	Binary Calculation	Decimal
a & b	1010 & 0011 = 0010	**2**
a | b	1010 | 0011 = 1011	**11**
a ^ b	1010 ^ 0011 = 1001	**9**
~a	~00001010 = 11110101	**-11** (- a+1)
a << 1	1010 -> 10100	**20** (2*a)
a >> 1	1010 -> 0101	**5** (a/2)

Output

Console Output

```
a & b:  2
a | b:  11
a ^ b:  9
~ a:  -11
a << 1:  20
a >> 1:  5
```

More Assignment Operators — Example Program

॥ सा विद्या या विमुक्तये ॥
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```c
#include <stdio.h>

int main()
{
    int a=10, b=3;

    a&=b;printf("value of a after a&=b: %d\n",a);
    a=10; a|=b;
    printf("value of a after a|=b: %d\n", a);
    a=10; a^=b;
    printf("value of a after a^=b: %d\n", a);
    a=10; a>>=1;
    printf("value of a after a>>=1: %d\n", a);
    a=10; a<<=1;
    printf("value of a after a<<=1: %d\n", a);

    return 0;
}
```

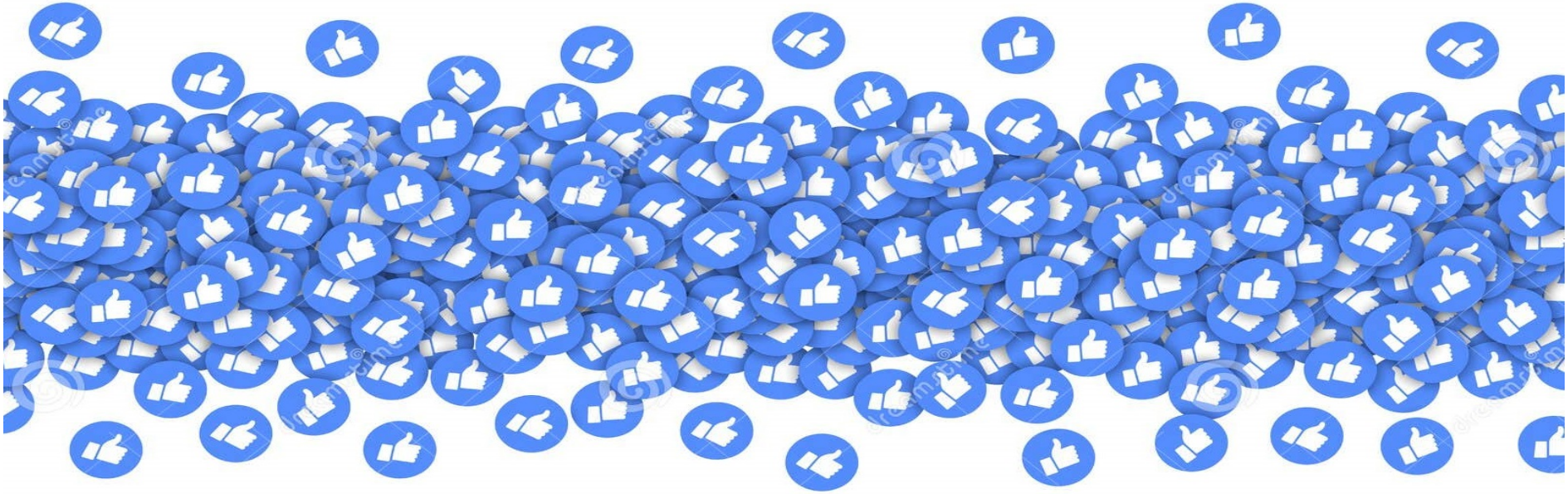
Output

Console Output

```
a & b:  2
a | b:  11
a ^ b:  9
a >> 1:  5
a << 1:  20
```

More Assignment Operators

Assignment	Equivalent to	Result
`a &= b`	`a = a & b`	**2**
`a |= b`	`a = a | b`	**11**
`a ^ = b`	`a = a ^ b`	**9**
`a <<= 1`	`a = a << 1`	**20**
`a >>= 1`	`a = a >> 1`	**5**



Thank you for your kind attention!!